

Alcovia Full Stack Engineering Intern Assignment - Master Guide

Best Project Names

1. SyncStudy (Recommended)
2. FocusSync
3. StudyFlow
4. AlcoSync
5. StudyMesh
6. FocusForge
7. EduSync
8. OfflineScholar
9. ConvergeStudy
10. SyncMind

Recommended final name: SyncStudy

What The Company Is Testing

Offline-first architecture, sync engine design, conflict resolution, idempotency, backend engineering, automation workflows, and system design judgment.

Features To Build

- A. Focus Sessions
 - Start session
 - Complete session
 - Give up
 - Offline support
 - Rewards, streaks, coins
- B. Syllabus Progress
 - Subjects
 - Chapters
 - Tasks
 - Progress rollups
- C. Automation Layer
 - Backend webhook
 - n8n workflow
 - Exactly-once notifications

Architecture

Frontend: React Native Expo + TypeScript
Backend: Express + TypeScript
Database: PostgreSQL
Local Storage: SQLite
Automation: n8n
Sync Model: Operation-based Sync
Conflict Resolution: Lamport Clock / Versioning
IDs: UUIDs

Database Design

students
focus_sessions
subjects
chapters
tasks
operations
processed_sessions
processed_events

Sync Approach

Store operations locally.
Sync only pending operations.
Server applies each operation exactly once.
Server returns latest state.
All devices eventually converge.

Conflict Resolution

Task edited on both devices:
Use Lamport Clock or Version-based merge.

Delete vs Update:
Delete wins using tombstones.

Duplicate sync:
Ignore duplicate operation IDs.

Out-of-order sync:
Lamport ordering handles sequencing.

Idempotency

Focus rewards counted exactly once.
Session IDs stored on server.
Duplicate sessions ignored.

n8n notifications use stable event IDs.
Duplicate notifications ignored.

Required Screens

1. Focus Session Screen
2. Syllabus Progress Screen
3. Dev Panel

Dev Panel:

- Online/Offline Toggle
- Sync Button
- Device State Viewer
- Pending Operations Viewer
- Notification Logs

Folder Structure

project/

frontend/
src/
screens/
components/
storage/
sync/
services/
hooks/
types/

backend/
src/
routes/
controllers/
services/
sync/
database/
middleware/

n8n/
workflow.json

README.md
DECISIONS.md

Deliverables

GitHub Repository
README.md
DECISIONS.md
n8n-workflow.json
5-minute demo video

Video Demo Checklist

Run two clients.
Take both offline.
Perform focus sessions.
Create conflicting task edits.
Reconnect.
Show reconciliation.
Show notification fires exactly once.
Explain sync strategy and limitations.

Recommended Implementation Plan

Day 1:
Database
Backend APIs
Local Storage
Sync Engine

Day 2:
Focus Sessions
Task Management

Conflict Resolution
n8n Integration
Dev Panel
README
DECISIONS.md
Video Recording